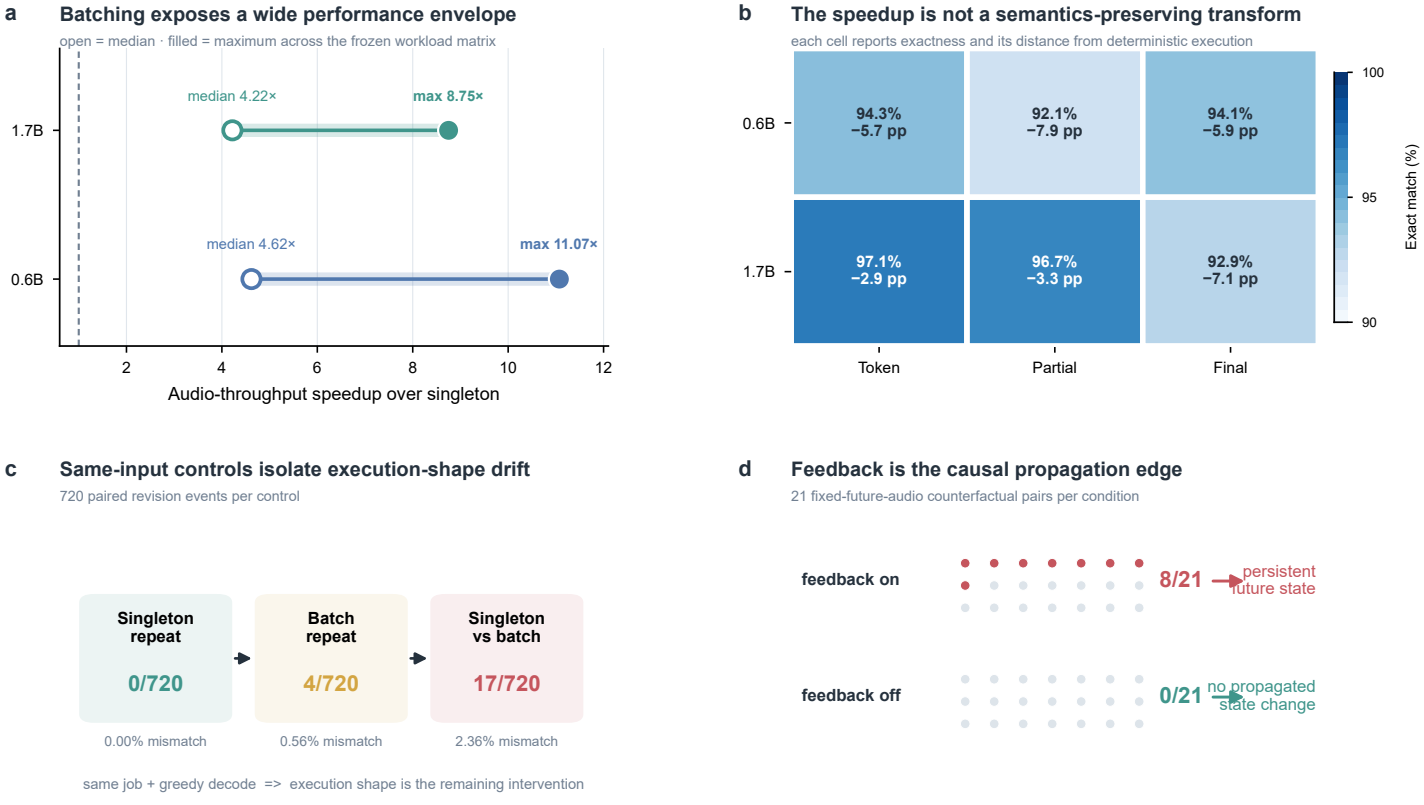

a
Batching exposes a wide performance envelope
open = median · filled = maximum across the frozen workload matrix
median 4.22×
max 8.75×
median 4.62×
max 11.07×
1.7B
0.6B
2
4
6
8
10
12
Audio-throughput speedup over singleton
b
The speedup is not a semantics-preserving transform
each cell reports exactness and its distance from deterministic execution
94.3%
−5.7 pp
92.1%
−7.9 pp
94.1%
−5.9 pp
97.1%
−2.9 pp
96.7%
−3.3 pp
92.9%
−7.1 pp
0.6B
1.7B
Token
Partial
Final
100
95
90
Exact match (%)
c
Same-input controls isolate execution-shape drift
720 paired revision events per control
Singleton repeat
0/720
0.00% mismatch
Batch repeat
4/720
0.56% mismatch
Singleton vs batch
17/720
2.36% mismatch
same job + greedy decode => execution shape is the remaining intervention
d
Feedback is the causal propagation edge
21 fixed-future-audio counterfactual pairs per condition
feedback on
8/21
persistent future state
feedback off
0/21
no propagated state change